Int. J. Production Economics 152 (2014) 57–70

Contents lists available at ScienceDirect

Int. J. Production Economics

journal homepage: www.elsevier.com/locate/ijpe

Modelling food logistics networks with emission considerations: The case of an international beef supply chain

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ARTICLE INFO

Article history:
Received 30 November 2012
Accepted 10 December 2013
Available online 19 December 2013

Keywords:
Multi-objective linear programming (MOLP)
Food logistics management
Livestock and beef
Sustainability
Carbon dioxide emissions
Perishability

ABSTRACT

Intrinsic characteristics of food products and processes along with growing sustainability concerns lead to the need for decision support tools that can integrate economic considerations with quality preservation and environmental protection in food supply chains. In this study, we develop a multi-objective linear programming (MOLP) model for a generic beef logistics network problem. The objectives of the model are (i) minimizing total logistics cost and (ii) minimizing total amount of greenhouse gas emissions from transportation operations. The model is solved with the ε-constraint method. This study breaks away from the literature on logistics network models by simultaneously considering transportation emissions (affected by road structure, vehicle and fuel types, weight loads of vehicles, traveled distances), return hauls and product perishability in a MOLP model. We present computational results and analysis based on an application of the model on a real-life international beef logistics chain operating in Nova Andradina, Mato Grosso do Sul, Brazil and exporting beef to the European Union. Trade-off relationships between multiple objectives are observed by the derived Pareto frontier that presents the cost of being sustainable from the point of reducing transportation emissions. The results from the pie chart analysis indicate the importance of distances between actors in terms of environmental impact. Moreover, sensitivity analysis on practically important parameters shows that export ports' capacities put pressure on the logistics system; decreasing fuel efficiency due to the bad infrastructure has negative effects on cost and emissions; and green tax incentives result in economic and environmental improvement.

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1. Introduction

The progressive increase of food consumption due to growing world population and wealth stimulates higher food production. A recent way for managing the increased production is globalization of food supply chains (FSCs) with the help of improvements in transport technologies, cheaper transportation, reductions in tariffs and other barriers to trade. Globalization has improved the chance of profitability from cross-border operations as well; however it has led to increased distances between partners in supply chains (Elhedhli and Merrick, 2012). The increased distances have enhanced the strategic importance of logistics network decisions such as selection of suppliers, distribution channels and transportation modes, determining production and inventory amounts at each plant and allocation of products (Cordeau et al., 2006; Harris et al., 2011). The need for a well organized logistics network thus has increased in the food sector, which is producing more than ever on a global scale.

Traditional logistics management considers mainly two key logistical aims, cost reduction (efficiency) and improved responsiveness, while dealing with the logistics network problem. However, intrinsic characteristics of food products and processes such as product perishability and food quality, and a growing sustainability trend require extension of the key logistical aims with quality and environmental considerations. This necessity leads to the need for decision support tools that can integrate economic considerations with quality preservation as well as environmental protection in FSCs. Accordingly, literature review shows that there is a need for models that are able to deal with the key challenges in managing quality and sustainability (Akkerman et al., 2010; Soysal et al., 2012). The need in practice and in research forms our main motivation to develop a model that allows to consider perishability of goods and emissions from transportation operations along with cost concerns in food logistics network.

We take the beef sector as a representative of a food supply chain that has both food and environment related challenges. Shelf life for beef that includes several quality factors (e.g. juiceness, tenderness, nutritive value, appearance and palatability) puts an additional pressure on logistics decisions, since the product may become undesirable, even it is not unsafe (Delmore, 2009). Apart

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http://dx.doi.org/10.1016/j.ijpe.2013.12.012

from quality concerns, appreciation grows for the idea of a carbon-constrained economy in the livestock sector with the growing awareness towards environment conservation (Robinson et al., 2011). Especially, transportation is one of the main sources of livestock related carbon dioxide (CO_2) emissions (Delgado et al., 1999) and increasing global beef trade results in more fuel consumption for beef related transportation. Therefore, it is wise to address product perishability and emissions from transportation while managing beef logistics chains.

We develop a multi-objective linear programming (MOLP) model for a generic beef logistics network problem. The objectives of the model are (i) minimizing total logistics cost and (ii) minimizing total amount of greenhouse gas (GHG) emissions from transportation operations. Duration of inventory keeping is limited due to the perishability nature of the product. The environmental effect of freight transportation is measured in CO_2 emissions. We provide a case study of the international beef logistics chain operating in Nova Andradina, Mato Grosso do Sul, Brazil and exporting beef to the European Union (EU) to illustrate the applicability of the proposed model for real logistics systems. The rationales for the selected beef chain are the following: (1) Brazil ranks as the largest beef exporter in the world by holding an approximately 21% share of the global beef trade in 2011 (Abiec, 2012d), (2) Brazil has potential to keep its position in the global market, and (3) Beef trade relationship exists (47,693 ton for fresh-chilled beef in 2010) between Brazil and the EU (Abiec, 2012a, 2012b). In this case study, we put the main focus on road transportation, which is the only delivery option till the export ports, as rail, inland ship or air need infrastructure that is not available yet. The logistical challenges in the case of Brazil are mainly related to usage of old trucks, inefficient road infrastructure or deficiency of available trucks.

The structure of the remaining paper is as follows. Section 2 presents a literature review on logistics models that take product perishability and/or emissions into account. Section 3 presents a formal definition of the generic problem, the methodology used for emission estimations and the proposed MOLP model for the generic beef logistics network problem. Section 4 presents the case study description, data gathering and the computational analysis of the model. The last section presents conclusions and directions for further research.

2. Literature review

The logistics network problem that has transportation and inventory decisions under capacity constraints for a multi-period planning horizon has been widely studied in the literature (see Ahn et al., 1994; Bilgen and Gunther, 2010; Verderame and Floudas, 2009). However, quality degradation of products puts additional challenges on logistics decisions in food sector. Literature review studies present the state of the art in product perishability consideration in FSCs (Ahumada and Villalobos, 2009; Akkerman et al., 2010; James et al., 2006; Soysal et al., 2012). As pointed out in these studies, the number of proposed decision support tools which are able to control products according to their quality levels has been increasing in recent years (e.g. Bosona and Gebresenbet, 2011; Rong et al., 2011; Wang et al., 2010; Ahuja, 2007). Ahuja (2007) controls the quality of products by constraining the number of periods that a good is stored at a facility. In this study, we use this approach to account for the perishable nature of beef.

Similar to raising awareness on quality decay, sustainability is an emerging area in FSCs (Akkerman et al., 2010; Seuring and Muller, 2008). The main reasons for the growing interest are stakeholder pressure and the need for adopting increasing environmental regulations. GHG emissions reduction, the most prominent environmental issue in practice, is one of the most significant sustainability objectives considered in logistics management literature (Soysal et al., 2012). Researchers have developed quantitative logistics models that can manage economic issues along with emission controls in response to the need for practice. Literature search is carried out within Thomson Reuters (formerly ISI) Web of Knowledge and followed by reference and citation analysis to find related contributions that have quantitative models with emission consideration for logistics management. We investigate the models with respect to main characteristics (Table 1) summarized below:

- *Model type*: Mixed Integer Linear Programming and Multi-Objective (Non)Linear Programming approaches are the most used modelling types.
- *Decisions*: The main logistical drivers in a supply chain are production/processing, transportation and inventory management decisions (Chopra and Meindl, 2010). The reviewed models manage one or more of the aforementioned decisions. All models aim to reduce emissions from transportation. Additionally, some studies consider emissions from production/processing and/or inventory holding together with transportation emissions.
- *GHG emissions calculation approaches:* The crucial stage during model development is calculating emissions from predetermined emission sources. Researchers employ basically two approaches to measure the emissions from transportation operations. First approach, which is preferred most, is using fixed emission or environmental impact factors per distance unit and/or per weight unit (e.g. Chaabane et al., 2008; Wang et al., 2011), per product (e.g. You et al., 2012), per vehicle (e.g. Paksoy et al., 2011b), which are obtained through other environmental studies. The second approach is estimating emissions indirectly by calculating total energy consumed from transportation operations while considering the aforementioned parameters such as distance, speed or weight (e.g. Bektas and Laporte, 2011; Bauer et al., 2010). For production and inventory related emissions, either fixed emission factor per unit produced or stocked (e.g. Oglethorpe, 2010) or energy consumption from production and/or inventory related operations is considered (e.g. Abdallah et al., 2012).
- *GHG consideration*: Studies in the literature either take only CO_2 gas emissions (e.g. Bauer et al., 2010) or group emissions of different GHG gases, such as CO_2, CH_4, and NO_x, together in a single indicator in terms of carbon dioxide equivalent (CO_2eq) emissions (e.g. You et al., 2012).
- *Application area*: Researchers implement the proposed models on different areas such as automotive, steel and plastic waste.

We found three quantitative models that manage product perishability while considering GHG emissions (Akkerman et al., 2009; Van der Vorst et al., 2009; You et al., 2012). Among these studies only You et al. (2012) propose a MOLP model that can be used to gain insight in the trade-off between multiple objectives. In contrast to that study, we also consider the effects of return hauls on transportation cost and emissions. Furthermore, we adopt a different methodology based on a distance-based formulation, Defra (2005), to estimate road transport emissions. Under this methodology, road structure, vehicle and fuel types, weight loads of vehicles and traveled distances are taken into account. This approach has been also used by Harris et al. (2011), who integrated the approach into a simulation model without considering perishability and return hauls (see Table 1). Therefore, this study breaks away from the literature on logistics network models by simultaneously considering the aforementioned issues.

Table 1
Logistics models that incorporate GHG emissions.

Studies	Model type	Decisions	GHG emissions calculation approaches	GHG consideration	Application area
Quariguasi-Frota-Neto et al. (2008)	MOLP	(PT)[a]	Environmental impact factor for producing one product unit and transporting one shipment of product is used.	U	Pulp and Paper
Akkerman et al. (2009)[b]	MILP	(PT)I	Environmental impact factor for producing one product unit and transporting one shipment of product is used.	CO_2eq	Meal elements
Van der Vorst et al. (2009)[b]	Sim.	P(TI)	Energy consumption for transportation and inventory is calculated and converted to realized emissions.	CO_2	Pineapples
Oglethorpe (2010)	GP	(PTI)	Emission factor per weight of output produced and stocked and per distance unit traveled is used. Energy used for refrigeration of product, machine use, heating and lightening is calculated and converted to realized emissions from inventory.	CO_2	Pork
Bauer et al. (2010)	MILP	(T)I	Energy consumption for transportation is calculated with a model and converted to realized emissions. The main parameters for the model are vehicle type and utilization ratio.	CO_2	U
Bektas and Laporte (2011)	ILP	(T)	Energy consumption for transportation is calculated with a model and converted to realized emissions. The main parameters for the model are: vehicle type, speed and utilization ratio	CO_2	U
Paksoy et al. (2011b)	MOLP	(T)	Emission factor per truck is used.	CO_2	U
Harris et al. (2011)	Sim.	(TI)	A formulation (Defra, 2005)[c] that estimates emissions from amount of fuel consumed is used. The effect of road structure on emissions is also considered. Electricity consumption in depots are estimated and converted to realized emissions from inventory.	CO_2	Automotive
Wang et al. (2011)	MOP-NL	P(T)	Emission factor per distance unit is used. Emissions from each unit of flow in facilities are also considered. There is possibility for making environmental investment on a facility to lower emissions.	CO_2	U
Chaabane et al. (2011)	MOLP	(PT)	Emission factor per weight unit and per distance unit is used.	CO_2eq	Steel
Paksoy et al. (2011a)	LP	P(T)	Emission factor per weight unit and per distance is used. Age of the vehicles is considered due to decreasing engine efficiency.	CO_2	U
Ubeda et al. (2011)	MILP	(T)	A formulation (Defra, 2005)[c] that estimates emissions from amount of fuel consumed is used. Amount of empty-running reduction during backhauls is also considered.	CO_2	Food distribution
Abdallah et al. (2012)	MILP	(PT)	Emission factor per weight unit and per distance unit are used. Energy consumption in facilities is also considered and converted to realized emissions.	CO_2	U
Bing et al. (2012)	MILP	(T)	Amount of fuel consumed is calculated and converted to realized emissions.	CO_2eq	Plastic waste
Chaabane et al. (2012)	MILP	(PT)I	Emission factor per output produced and distributed is used.	CO_2eq	Aluminium
Elhedhli and Merrick (2012)	MIP-NL	P(T)	Emission factor per vehicle is used. The relationship between emissions and vehicle weight is modeled using a concave function.	CO_2	U
Mallidis et al. (2012)	MILP	(T)	Emission factor per weight unit and per distance are used.	CO_2eq	U
You et al. (2012)[b]	MOLP	(PTI)	Emission factor per output produced, distributed and stored is used. Life Cycle Assessment methodology is employed for identifying and quantifying the emissions.	CO_2eq	Cellulosic, ethanol sector

P: Production/Processing, T: Transportation, I: Inventory, CO_2: Carbon dioxide, CO_2eq: Carbon dioxide equivalent, MILP: Mixed-integer linear programming, MOP-NL: Non-linear multi-objective programming, Sim: Simulation, GP: Goal programming, MOLP: Multi-objective linear programming, ILP: Integer linear programming, LP: Linear programming, MIP-NL: Non-linear mixed integer programming, U: Unspecified.

[a] Parentheses refer emission source, for instance (P) refers emissions from production or (PT) refers emissions from production and transportation.
[b] Those studies have product perishability consideration.
[c] Detailed information about this model is given in coming section.

Consequently, contributions of the study can be summarized as follows: (1) integrating food transport emissions into a MOLP model for the generic beef logistics network problem while considering road structure, vehicle and fuel types, weight loads of vehicles, traveled distances, return hauls and product perishability, (2) presenting the applicability of the model in an international beef logistics network based on real data, multiple scenarios, and analysis.

3. Modelling a generic beef logistics network

3.1. Formal definition of the generic problem

Our modelling approach is based on a generic multi-echelon beef logistics network problem that consists of a number of third party logistics (3PL) firms, production regions, slaughterhouses, export departure and import arrival points at fixed locations (Fig. 1). In the generic beef logistics chain, production regions are responsible for providing livestock to slaughterhouses. Slaughterhouses can be supplied from more than one production region and after the slaughtering process, beef can be sent to more than one export departure point from the same slaughterhouse. Slaughterhouses can keep limited amount of livestock inventory. Different types of rented trucks from 3PL firms are used for livestock transportation between production regions and slaughterhouses, and for beef transportation between slaughterhouses and export departure points. Each truck rented from a 3PL firm turns back again to the same firm (see Fig. 1) and trucks do not need to be fully loaded during service. It is also worth to mention that trucks are empty before getting to the sites for service and during return stage to the 3PL firms. Transportation capacity restrictions are imposed on export departure points. There are multiple transport options such as sea, train or air transportation, between export departure and import arrival points. Import arrival points have also a multi-source option that allows receiving beef from different export departure points.

The demand for beef is assumed to be known a priori. CO_2 emissions occur due to the transportation between the actors. Slaughterhouses and export departure points can keep certain amounts of beef inventory. However, it is not possible to keep long term beef inventory in those facilities due to quality considerations.

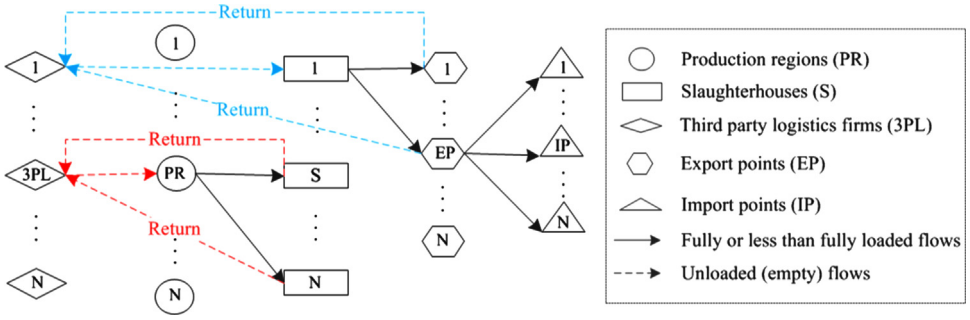


Fig. 1. Representation of the generic beef logistics network.

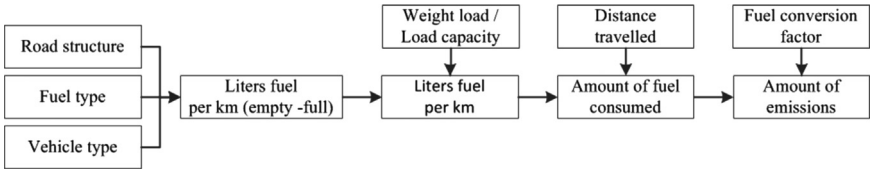


Fig. 2. Emissions estimation methodology for road transportation.

Therefore, maximum number of periods that beef can be stored in facilities needs to be restricted. Additionally, indirect flows between actors are not allowed. That means each facility sends livestock or beef directly to its destination location.

The described generic problem allows us to present the integration of emissions generated from beef and livestock transportation into a mathematical logistics network model. We acknowledge that different beef logistics networks can be characterized in several ways in terms of product flows, truck flows and ownership or even number of chain layers. However, it is possible to employ the same emission estimation methodology, which is described in Section 3.2, and same approach to incorporate emissions into the model, which is described in Section 3.4, while adapting the model to different logistics structures.

The decisions that need to be made are (i) number of livestock slaughtered per slaughterhouse per period, (ii) amount of livestock and beef inventories, (iii) flows between actors (allocation decisions), (iv) number of trucks used (rented) from 3PL firms considering also the possibility of less than fully loaded truck shipments. The planning horizon comprises N periods. Transportation lead time between the actors in the chain is assumed to be zero; however, it can be easily incorporated into the model. The aim of the model is minimizing total logistics costs comprising inventory and transportation costs together with the total CO_2 emissions from transportation operations.

3.2. Emissions estimations

We follow the methodology presented in Fig. 2 to calculate CO_2 emissions from road transportation, which is based on a distance-based formulation (Defra, 2005, 2011). The main required parameters for the formulation are liters fuel per km (*Lfpk*) for empty and full trips. Vehicle and fuel type, and road structure affect these parameters. The distance-based formulation assumes that load linearly affects fuel consumption. Therefore, we calculate *Lfpk* for trucks by considering their load weights, and *Lfpk* for empty and full trips as follows (Defra, 2005, 2011):

$$Lfpk = Lfpk(empty) + (Lfpk(full) - Lfpk(empty)) * (weight\ load/load\ capacity)$$

Then, multiplication of *Lfpk* with traveled distances gives us total fuel consumption amounts. After calculating total fuel consumption amounts, we estimate CO_2 emissions by taking into account the amount fuel consumed and fuel conversion factor as follows (Defra, 2005, 2011):

$$Total CO_2(kg) = Amount\ of\ fuel\ consumed\ (Total\ km\ travelled * Lfpk) * Fuel\ conversion\ factor$$

This approach allows us to consider the effects of empty drives and less than fully loaded truck shipments on logistics cost and CO_2 emissions. We use unit CO_2 emissions (kg/ton km) to calculate emissions from other transportation modes such as sea, train or air transportation.

3.3. Notation

For the mathematical description of the model the following notation is introduced:

Indices

i	index for facilities including production regions, slaughterhouses, export departure and import arrival points
k	node index for 3PL firms
i,j	index pair referring to an arc from node i to node j
m	vehicle type index
f	transportation mode index
t	time index referring to a period

Sets

F	set of production regions
S	set of slaughterhouses
P	set of export departure points
C	set of import arrival points
FS	set of production regions and slaughterhouses
FS_a	set of arcs between production regions and slaughterhouses
SP	set of slaughterhouses and export departure points
SP_a	set of arcs between slaughterhouses and export departure points
PC	set of export departure points and import arrival points
PC_a	set of arcs between export departure points and import arrival points
O	set of 3PL firms
OF_a	set of arcs between 3PL firms and production regions

OS_a set of arcs between 3PL firms and slaughterhouses
OP_a set of arcs between 3PL firms and export departure points
$M_{i,j}$ set of truck types that can be used for arc $i,j \in FS_a \cup SP_a$
$TM_{i,j}$ set of transportation modes that can be used for arc $i,j \in PC_a$
T set of periods and L refers the length of a future planning cycle

Monetary parameters

$liveinvcost_i$ cost for storing one livestock unit for one period in facility $i \in S$
$beefinvcost_i$ cost for storing one ton beef (bone free meat) for one period in facility $i \in SP$
$rentcost_{k,i,j,m}$ fixed renting cost from 3PL firm $k \in O$, to use in arc $(i,j) \in FS_a \cup SP_a$, for truck type $m \in M_{i,j}$
$fuelcost$ fuel cost per liter
$unitcost_{i,j,f}$ cost per ton for transportation on arc $(i,j) \in PC_a$ with transportation mode $f \in TM_{i,j}$

Technical parameters

max maximum number of periods that beef can be stored in facilities, max $\leq L-1$
$weight$ average carcass weight of one livestock, in tons
$yield$ yield of one livestock, 1 ton carcass weight = 0.7 (yield ratio) ton beef
$emptyfuel_{i,j,m}$ the liter fuel consumption amount (l/km) of empty truck in arc $(i,j) \in FS_a \cup SP_a \cup OF_a \cup OS_a \cup OP_a$ with truck type $m \in M_{i,j}$
$fullfuel_{i,j,m}$ the liter fuel consumption amount (l/km) of full truck in arc $(i,j) \in FS_a \cup SP_a \cup OF_a \cup OS_a \cup OP_a$, with truck type $m \in M_{i,j}$
$conversion$ fuel conversion factor
$capacity_m$ transportation capacity of truck type $m \in M_{i,j}$, in tons/heads
$distance_{i,j}$ distance between node $i \in FS \cup P \cup O$ and node $j \in FS \cup P \cup O$, in km
$demand_{i,t}$ demand from import arrival point $i \in C$, in period t, in tons
$livecap_{i,t}$ number of available livestock in production region $i \in F$, at the beginning of period $t \in T$, in tons
$liveinvcap_i$ total livestock inventory capacity of slaughterhouse $i \in S$ for the whole planning horizon, in tons
$slaughtercap_i$ total livestock slaughtering capacity of slaughterhouse $i \in S$ for the whole planning horizon, in tons
$beefinvcap_i$ total beef inventory capacity of facility $i \in SP$ for the whole planning horizon, in tons
$transcap_i$ total beef transportation capacity of export departure point $i \in P$ for the whole planning horizon, in tons
$emissions_{i,j,f}$ CO_2 emissions factor (kg/ton km) for transportation on arc $(i,j) \in PC_a$ with transportation mode $f \in TM_{i,j}$, in kg/ton km

Main decision variables

$L_{i,j,t,m}$ flow quantities of livestock on arcs $(i,j) \in FS_a$, in period $t \in T$, with truck type $m \in M_{i,j}$, in number of heads
$BT_{i,j,t,m}$ flow quantities of beef on arcs $(i,j) \in FS_a \cup SP_a$, in period $t \in T$, with truck type $m \in M_{i,j}$, in tons
$BS_{i,j,t,f}$ flow quantities of beef on arcs $(i,j) \in PC_a$, in period $t \in T$, with transportation mode $f \in TM_{i,j}$, in tons
$N_{k,i,j,t,m}$ number of fully loaded trucks rented from 3PL firm $k \in O$, used on arcs $(i,j) \in FS_a \cup SP_a$, in period t, with truck type $m \in M_{i,j}$
$C_{i,t}$ number of livestock slaughtered in slaughterhouse $i \in S$, in period $t \in T$
$IL_{i,t}$ inventory level of livestock in slaughterhouse $i \in S$, at the beginning of period $t \in T$, in number of heads
$IB_{i,t}$ inventory level of beef in facility $i \in SP$, at the beginning of period $t \in T$, in tons
$LT_{k,i,j,t,m}$ amount of load carried with less than fully loaded truck rented from 3PL firm $k \in O$, on arcs $(i,j) \in FS_a \cup SP_a$, in period $t \in T$, with truck type $m \in M_{i,j}$, in number of heads or tons

Derived decision variables

$Z_{k,i,j,t,m}$ binary variable that equals 1, if less than full truck load is carried with a truck rented from 3PL firm $k \in O$, on arc $(i,j) \in FS_a \cup SP_a$, in period t, with truck type $m \in M_{i,j}$, otherwise 0
$U_{k,i,j,t,m}$ utilization rate (load factor) of less than fully loaded truck rented from 3PL firm $k \in O$, used on arc $(i,j) \in FS_a \cup SP_a$, in period $t \in T$, with truck type $m \in M_{i,j}$
$LF_{k,i,j,t,m}$ the liter fuel consumption amount (l/km) of less than fully loaded truck rented from 3PL firm $k \in O$, used on arc $(i,j) \in FS_a \cup SP_a$, in period $t \in T$, with truck type $m \in M_{i,j}$

3.4. Multi-objective linear programming (MOLP) model for the generic beef logistics network problem

The generic logistics network problem is formulated mathematically as a MOLP model. The objectives of the model are (i) an economic objective to minimize total logistics cost and (ii) an environmental objective to minimize total CO_2 emissions from transportation operations. The model constraints include the following: inventory and product flow balance, demand satisfaction, flow structure, truck utilization rate, fuel consumption amount, capacity of transport, inventory and slaughtering, and decision variable constraints.

Economic objective (OF_1): The economic objective is measured by the total logistics cost that consists of the sum of four parts:

$$\textbf{min } OF_1 = IC + TC_1 + TC_2 + TC_3 \quad (1)$$

- Inventory costs (IC) for livestock and beef:

$$IC = \sum_{t=1}^{L} \sum_{i \in S} liveinvcost_i * IL_{i,t} + \sum_{t=1}^{L} \sum_{i \in SP} beefinvcost_i * IB_{i,t}$$

- Transportation costs of fully loaded trucks (TC_1) considering also empty arrivals to sites and returns to 3PL firms:

$$TC_1 = \sum_{t=1}^{L} \sum_{(i,j) \in FS_a \cup SP_a} \sum_{m \in M_{i,j}} \sum_{k \in O} (N_{k,i,j,t,m} * (rentcost_{k,i,j,m} + (distance_{i,j} * fullfuel_{i,j,m} * fuelcost + distance_{k,i} * emptyfuel_{k,i,m} * fuelcost + distance_{j,k} * emptyfuel_{j,k,m} * fuelcost)))$$

- Transportation costs of less than fully loaded trucks (TC_2) considering also empty arrivals to sites and returns to 3PL firms:

$$TC_2 = \sum_{t=1}^{L} \sum_{(i,j) \in FS_a \cup SP_a} \sum_{m \in M_{i,j}} \sum_{k \in O} ((Z_{k,i,j,t,m} * rentcost_{k,i,j,m}) + (distance_{i,j} * LF_{k,i,j,t,m} * fuelcost) + Z_{k,i,j,t,m} * (distance_{k,i} * emptyfuel_{k,i,m} * fuelcost + distance_{j,k} * emptyfuel_{j,k,m} * fuelcost))$$

- Transportation costs of other transportation modes such as sea, train or air (TC_3) between export departure and import arrival points:

$$TC_3 = \sum_{t=1}^{L} \sum_{(i,j) \in PC_a} \sum_{f \in TM_{i,j}} BS_{i,j,t,f} * unitcost_{i,j,f}$$

Environmental Objective (OF_2): The environmental objective is measured by the total CO_2 emissions that consists of the sum of three parts:

$$\textbf{min } OF_2 = TE_1 + TE_2 + TE_3 \tag{2}$$

- Transportation emissions from fully loaded trucks (TE_1) considering also empty arrivals to sites and returns to 3PL firms:

$$TE_1 = \sum_{t=1}^{L} \sum_{(i,j) \in FS_a \cup SP_a} \sum_{m \in M_{i,j}} \sum_{k \in O} (N_{k,i,j,t,m} * (distance_{i,j} * fullfuel_{i,j,m} * conversion + distance_{k,i} * emptyfuel_{k,i,m} * conversion + distance_{j,k} * emptyfuel_{j,k,m} * conversion))$$

- Transportation emissions from less than fully loaded trucks (TE_2) considering also empty arrivals to sites and returns to 3PL firms:

$$TE_2 = \sum_{t=1}^{L} \sum_{(i,j) \in FS_a \cup SP_a} \sum_{m \in M_{i,j}} \sum_{k \in O} (distance_{i,j} * LF_{k,i,j,t,m} * conversion + Z_{k,i,j,t,m} * (distance_{k,i} * emptyfuel_{k,i,m} * conversion + distance_{j,k} * emptyfuel_{j,k,m} * conversion))$$

- Transportation emissions from other transportation modes such as sea, train or air (TE_3) between export departure and import arrival points:

$$TE_3 = \sum_{t=1}^{L} \sum_{(i,j) \in PC_a} \sum_{f \in TM_{i,j}} BS_{i,j,t,f} * emissions_{i,j,f} * distance_{i,j}$$

Constraints: The model consists of the following sets of constraints:

- Constraints (3) and (4) ensure capacity restrictions on livestock transportation from production regions to slaughterhouses and balanced livestock inventories in slaughterhouses:

$$\sum_{j \in S} \sum_{m \in M_{i,j}} L_{i,j,t,m} \le livecap_{i,t}, \quad \forall i \in F, \ \forall t \in T, \tag{3}$$

$$IL_{i,t} + \sum_{j \in F} \sum_{m \in M_{i,j}} L_{j,i,t,m} - C_{i,t} = IL_{i,t+1}, \quad \forall i \in S, \ \forall t \in T, \tag{4}$$

- Constraints (5)–(8) ensure balanced beef inventories in slaughterhouses and export departure points while adhering to maximum storage time constraint of beef:

$$IB_{i,t} + (C_{i,t} * weight * yield) - \sum_{j \in P} \sum_{m \in M_{i,j}} BT_{i,j,t,m} = IB_{i,t+1}, \quad \forall i \in S, \ \forall t \in T, \tag{5}$$

$$IB_{i,t} \le \sum_{a=t}^{t+max} \sum_{j \in P} \sum_{m \in M_{i,j}} BT_{i,j,a,m}, \quad \forall i \in S, \ \forall t \in T, \tag{6}$$

$$IB_{i,t} + \sum_{j \in S} \sum_{m \in M_{i,j}} BT_{j,i,t,m} - \sum_{j \in C} \sum_{f \in TM_{i,j}} BS_{i,j,t,f} = IB_{i,t+1}, \quad \forall i \in P, \ \forall t \in T, \tag{7}$$

$$IB_{i,t} \le \sum_{a=t}^{t+max} \sum_{j \in C} \sum_{f \in TM_{i,j}} BS_{i,j,a,f}, \quad \forall i \in P, \ \forall t \in T, \tag{8}$$

- Constraints (9) enforce that the total rate of flow from the export departure points must be higher than the corresponding market demand of import arrival points:

$$\sum_{j \in P} \sum_{f \in TM_{i,j}} BS_{j,i,t,f} \ge demand_{i,t}, \quad \forall i \in C, \forall t \in T, \tag{9}$$

- Constraints (10)–(13) represent that flows between nodes comprise fully and less than fully loaded truck trips. However, only one less than fully loaded truck can be on an arc at the same period and truck load needs to be less than truck capacity:

$$L_{i,j,t,m} = \sum_{k \in O} N_{k,i,j,t,m} * capacity_m + LT_{k,i,j,t,m}, \quad \forall (i,j) \in FS_a, \ \forall t \in T, \ \forall m \in M_{i,j}, \tag{10}$$

$$BT_{i,j,t,m} = \sum_{k \in O} N_{k,i,j,t,m} * capacity_m + LT_{k,i,j,t,m}, \quad \forall (i,j) \in SP_a, \ \forall t \in T, \ \forall m \in M_{i,j}, \tag{11}$$

$$\sum_{k \in O} \sum_{m \in M_{i,j}} Z_{k,i,j,t,m} \le 1, \quad \forall (i,j) \in FS_a \cup SP_a, \ \forall t \in T, \tag{12}$$

$$LT_{k,i,j,t,m} < capacity_m * Z_{k,i,j,t,m}, \quad \forall (i,j) \in FS_a \cup SP_a, \ \forall t \in T, \ \forall m \in M_{i,j}, \ \forall k \in O, \tag{13}$$

- Constraints (14) ensure that utilization rates of less than fully loaded trucks are calculated based on less than full truck loads and capacities:

$$U_{k,i,j,t,m} = LT_{k,i,j,t,m} / capacity_m, \quad \forall (i,j) \in FS_a \cup SP_a, \ \forall t \in T, \ \forall m \in M_{i,j}, \ \forall k \in O, \tag{14}$$

- Constraints (15) ensure that the liter fuel consumption amount (l/km) of less than fully loaded trucks are calculated based on the distance based formulation (Defra, 2005, 2011) described in Section 3.2:

$$LF_{k,i,j,t,m} = (Z_{k,i,j,t,m} * emptyfuel_{i,j,m}) + ((fullfuel_{i,j,m} - emptyfuel_{i,j,m}) * U_{k,i,j,t,m}), \forall (i,j) \in FS_a \cup SP_a, \forall t \in T, \forall m \in M_{i,j}, \forall k \in O, \tag{15}$$

- Constraints (16)–(19) ensure capacity restrictions on livestock slaughtering, livestock and beef stocking, and beef transportation to ports:

$$\sum_{t=1}^{L} C_{i,t} \le slaughtercap_i, \quad \forall i \in S, \tag{16}$$

$$\sum_{t=1}^{L} IL_{i,t} \le liveinvcap_i, \quad \forall i \in S, \tag{17}$$

$$\sum_{t=1}^{L} IB_{i,t} \le beefinvcap_i, \quad \forall i \in SP, \tag{18}$$

$$\sum_{t=1}^{L} \sum_{j \in S} \sum_{m \in M_{i,j}} BT_{j,i,t,m} \le transcap_i, \quad \forall i \in P, \tag{19}$$

- Constraints (20)–(26) represent the nonnegativity, integrality and binary restrictions imposed upon the decision variables:

$$L_{i,j,t,m} \ge 0, \quad \forall (i,j) \in FS_a, \ \forall t \in T, \ \forall m \in M_{i,j}, \tag{20}$$

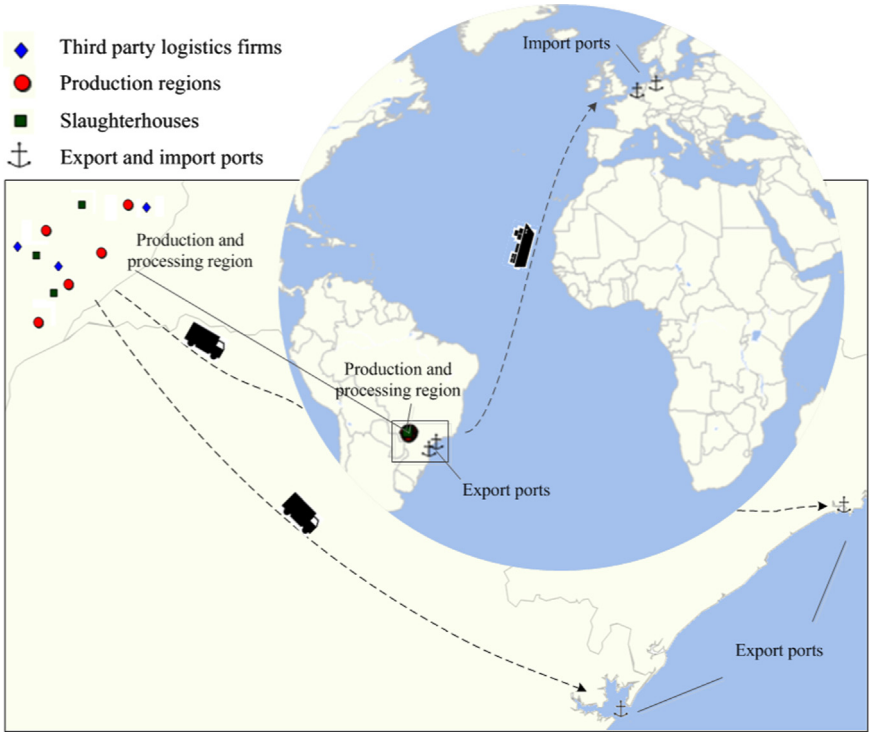


Fig. 3. The analysed beef logistics network between Nova Andradina and the EU.

$$BT_{i,j,t,m} \geq 0, \quad \forall (i,j) \in SP_a, \quad \forall t \in T, \quad \forall m \in M_{i,j}, \tag{21}$$

$$BS_{i,j,t,f} \geq 0, \quad \forall (i,j) \in PC_a, \quad \forall t \in T, \quad \forall f \in TM_{i,j}, \tag{22}$$

$$C_{i,t}, IL_{i,t} \geq 0, \quad \forall i \in S, \quad \forall t \in T, \tag{23}$$

$$IB_{i,t} \geq 0, \quad \forall i \in SP, \quad \forall t \in T, \tag{24}$$

$$LT_{k,i,j,t,m}, U_{k,i,j,t,m}, LF_{k,i,j,t,m} \geq 0, \quad \forall (i,j) \in FS_a \cup SP_a, \quad \forall t \in T, \quad \forall m \in M_{i,j}, \quad \forall k \in O, \tag{25}$$

$$N_{k,i,j,t,m} \in \mathbb{Z}^+, Z_{k,i,j,t,m} \in \{0,1\}, \quad \forall (i,j) \in FS_a \cup SP_a, \quad \forall t \in T, \quad \forall m \in M_{i,j}, \quad \forall k \in O, \tag{26}$$

4. Case study

4.1. Description and data gathering

This section presents an implementation of the proposed model in a real-life fresh-chilled beef logistics network operating in Nova Andradina, Mato Grosso do Sul, Brazil and exporting beef to the EU. The overview of the whole network and the zoomed view of the area that covers production regions, 3PL firms, slaughterhouses and export ports are presented in Fig. 3. Mato Grosso do Sul ranks third out of 26 states in terms of contribution (11.74%) to total area used for cattle in hectares in Brazil (Nzte, 2010). Mato Grosso do Sul has 11 micro-regions. The reasons for selecting Nova Andradina among other micro-regions are the following: (1) it has three slaughterhouses that have certification to export cattle meat to EU out of 11 in Mato Grosso do Sul and (2) it has a high density of cattle production. Nova Andradina has five cities (Table A3). We consider each city as a production region, since there are hundreds of small and medium sized farms in the region. We have a planning horizon of 6 months and use the first 6 months of 2010s trade data between Brazil and EU in our analysis (Abiec, 2012a, 2012b; Nzte, 2010).

Fleet age is one of the major issues for logistics management, since age of the vehicles apparently has an important effect on fuel consumptions because of new technologies. The average fleet age used in logistics activities is more than 8 years in Brazil (World-Bank, 2012). Due to this fact, rather than considering different types of vehicles used, we prefer to consider the issue of fleet ages that can be also handled with our model. The difference in the model is that $M_{i,j}$, set of truck types, is interpreted as set of truck age categories and m, vehicle type indices, are interpreted as vehicle age category indices. The fleet in Brazil includes tractor-trailers, fixed bed, bulk goods, and special goods vehicles such as tankers (World-Bank, 2012). We assume that livestock and beef are carried to the slaughterhouses and export ports with standard trucks (Tractor semi-trailers) that have two age categories (old and new) and rented from one of the three 3PL firms in Nova Andradina. Transportation capacities of trucks are 20 ton beef and 20 cattle (Cederberg et al., 2009).

We calculate cost of operations of trucks by considering fixed renting costs (€/truck) from the 3PL firms and variable costs (€/km) over distances. We estimate fixed renting costs for trucks considering related capital investment of 3PL firms depending on the age of the truck. So, it is a reasonable assumption to take fixed renting costs for the new truck higher than the old truck. Basically, two stages exist in the beef logistics network: 1st stage consisting all routes that start from a 3PL firm, visit a production region and a slaughterhouse and turn back to the same 3PL firm; 2nd stage consisting all routes that start from a 3PL firm, visit a slaughterhouse and an export port and turn back to the same 3PL firm. Therefore, fixed costs cover the period that starts when a truck departs from a 3PL firm and ends when the empty truck returns to the 3PL firm after finalizing the service. Fixed renting costs (240 €/old truck, 420 €/new truck) for the routes in 2nd stage are higher than the costs (40 €/old truck, 70 €/new truck) for the routes in 1st stage because of the distance differences (see Fig. 3).

Table 2
Fully loaded and unloaded (empty) fuel consumption amounts (l/km) calculation.
Source: Based upon the study of Hoen et al. (2010).

Road type	Motorway		Rural		Urban		3PL-PR/PR-S/S-3PL		S-EP/EP-3PL	
Load factor (%)	**0**	**100**	**0**	**100**	**0**	**100**	**0**	**100**	**0**	**100**
Base case	0.226	0.360	0.230	0.396	0.288	0.504	–	–	–	–
Old vehicle (+10%)	0.249	0.396	0.253	0.436	0.317	0.554	**0.258**	**0.432**	**0.263**	**0.432**
New vehicle (−10%)	0.203	0.324	0.207	0.356	0.259	0.454	**0.211**	**0.353**	**0.215**	**0.353**

3PL: Third party logistics firms, PR: Production regions, S: Slaughterhouses, EP: Export Ports.

Unfortunately, no data is available on Brazilian fuel consumption amounts of trucks for different road types. Therefore, we use the data given by Hoen et al. (2010) for the trucks in Europe's conditions. The data is presented in Table 2 as a base case. We assume that the data of Hoen et al. (2010) relate to refrigerated trucks and an old truck consumes 10% higher and a new truck consumes 10% lower than the base case amounts (Table 2). It is assumed that road structures of 3PL-production regions, production regions-slaughterhouses and slaughterhouses-3PL are the same: 40% motorway, 50% rural and 10% urban. Road structures of slaughterhouses-export ports, and export ports-3PL are also the same: 70% motorway, 10% rural and 20% urban. Related fuel consumption amounts between actors based on the aforementioned road structures are presented in Table 2. We use 2.63 kg/l for fuel conversion factor (Defra, 2007) and €1 for fuel cost in Brazil (http://data.worldbank.org, 2012) to calculate emissions and operational cost.

Inventory holding costs for livestock (22.05€/head month) and beef (105€/ton month) are expressed as a percentage (3%) of the approximate market values of the items and same for all facilities. We take the market price for one ton bone-free beef as €3500 (www.swineweb.com, 2012), average carcass weight of one livestock as 300 kg and use the factor 0.7, i.e. from 1 kg carcass weight (meat with bone), 0.7 kg bone-free beef is produced (Cederberg et al., 2009). Correspondingly, inventory costs are calculated as follows: (i) livestock inventory cost per head: Market price * Carcass weight * Yield ratio * Percentage (3%) and (ii) beef inventory cost per ton: Market price * Percentage (3%).

The Nova Andradina region uses Porto de Paranagua and Santos Ports for beef export to EU ports which are in Rotterdam and Hamburg. Unit ton cost for transporting between ports with ship (PS-type container vessel: 11,000 TEU) is taken as 0.12 €/kg (www.globalshippingcosts.com, 2012). Related CO_2 emissions factor is 0.007 kg/ton km (Dekker et al., 2012). EU share in Brazil exports is 5% for fresh-chilled beef (Abiec, 2012b). State of Mato Grosso do Sul's participation is taken as 11.74% (Nzte, 2010). Finally, Nova Andradina's contribution to total export is estimated as 27% (# of export slaughterhouses in Nova Andradina /# of export slaughterhouses in Mato Grosso do Sul). Monthly European demand satisfied from Nova Andradina for fresh-chilled beef in the first 6 months of 2010 is presented in Table A1 and it is assumed that demand of EU is satisfied equally from the two import ports, Rotterdam and Hamburg.

Distances between actors (Table A2) are calculated via web based distance measurement tools (www.maps.google.nl, 2012; www.searates.com, 2012). Locations of production regions are defined as approximately center of each city. Available livestock numbers in production regions are estimated considering their contributions to cattle production in 2010 (Table A3) and the demand satisfied from the region (Table A1). Slaughtering capacities (300 ton) are taken as the same for all slaughterhouses and estimated considering the demand satisfied from the region. Another assumption is that slaughterhouses can keep livestock (300 head) and beef (200 ton) inventory, whereas export ports can keep only beef (100 ton) inventory. Maximum storage time, k, is restricted to 2 months because of perishability nature of the beef. Export ports' transportation capacities used by Nova Andradina region are estimated (Porto de Paranagua, 124 tons, Santos Port, 676 tons) considering the amounts of Brazilian beef exports by sea ports in 2010 (Abiec, 2012c). Summary for all the values and estimations of the input parameters along with the relevant data sources are presented in Table A4.

4.2. Model solution

We solved the MOLP model with the ε-constraint method (Andersson, 1999). This method has been also employed in other recent studies which have multiple objective models (Chaabane et al., 2011; Nikbakhsh et al., 2013; Zhang et al., 2010). Under the ε-constraint method, one objective is selected for optimization, whereas the others are reformulated as constraints. In our solution methodology, OF_1 was selected for optimization and OF_2 was formulated as an additional constraint. The right hand side value of the additional constraint is ε, which represents the limit on CO_2 emissions. We derived a Pareto frontier to observe the dependency between the two objectives. While deriving the Pareto frontier, initially, the model that has OF_1 was solved without the additional constraint and the total amount of generated emissions in that instance was set as a highest value of the ε. Afterwards, the additional constraint was activated and the right hand side value of the ε was progressively reduced in each instance starting from its highest value. Therefore, a set of models that differ with respect to the right hand side value of the additional constraints is required. Consequently, progressively changing the ε value allowed us to obtain different points on the Pareto frontier:

$$\textbf{min} \quad OF_1$$

s.t.

Constraints (3)–(26),

$$OF_2 \le \varepsilon, \quad \text{(additional constraint)}$$

We used the ILOG-OPL development studio and Cplex 12.2 optimization solver to solve the MOLP model. The model has 2637 continuous, 756 binary and 756 integer variables, and 2806 constraints. We could not find optimal solutions within a time limit of 1000 s because of the size of the problem. However, we calculated lower bounds for each model through a relaxation technique, where all binary variables related with less than fully loaded truck usage were replaced by continuous variables, constrained in the [0, 1] interval. Our computational tests showed that the average difference between our feasible solutions and the lower bounds is less than approximately 2%. Therefore, we used the feasible solutions in our analysis that are sufficiently close to the optimal solutions and obtained within reasonable times.

4.3. Analysis and discussion

4.3.1. Solutions of base cases

We defined two base cases: Lowest Cost (LC) and Lowest emission (LE). Summary results for these base cases are presented in Table 3. The effect of reducing emissions can be seen from the total cost difference between LC and LE base cases. Inventory and road transportation cost items increase in LE base case. The reasons for the cost change in LE base case are (i) decreasing the number of less than fully loaded trucks that result an increase in inventories, and (ii) using new trucks rather than old ones that result an increase in transportation costs.

Emission differences in different parts of the logistics network in both cases are due to road conditions, distances and vehicle loads. Ship transportation related cost and emissions do not change between the base cases, since it is the single alternative for transportation between export and import ports and age of the vehicles for ship transportation is not considered in our problem.

Results suggest to keep 55 and 5.2 tons of beef inventory for the whole planning horizon in the LE and LC cases respectively. Although maximum storage time is restricted to 2 months, under the suggested strategy beef would not be held in inventory longer than 1 month (Table 3). This indicates that constraints (6) and (8) are non-binding constraints as a result of the parameter setting of our case study. However, these two constraints will restrict the maximum storage time for FSCs where keeping inventory more than 1 month is advantageous.

The structure of the analysed logistics network and the aggregated values (sum of all planning horizon) of the decision variables for the LC case are visualized in Fig. 4. In order to not complicate the figure, the returns from slaughterhouses and export ports to 3PL firms, and the number of less than fully loaded trucks in each route are not shown. However, regarding the returns, it is already known that each truck rented from a 3PL firm turns back again to the same firm (see Fig. 1) due to the given assumption. The total beef inventory for the whole planning horizon (5.2 tons), which is also not presented in the figure, is kept in the Santos Port.

Table 3
Summary results for base cases.

Case	Lowest cost	Lowest emission
Logistics cost (€)		
Inventory	546	7004
Road transportation	44,642	51,801
Ship transportation	93,720	93,720
Total	138,908	152,525
Transportation emissions (kg)		
Third party logistics firms–production regions	4000	3100
Production regions–slaughterhouses	7196	5554
Slaughterhouses–third party logistics firms	5160	4264
Slaughterhouses–export ports	34,514	27,856
Export ports–third party logistics firms	20,672	16,485
Export ports–import ports	56,374	56,374
Total	127,917	113,633
General		
Less than fully loaded trucks (%)	12	6
Old truck usage (%)	100	0
New truck usage (%)	0	100
Total livestock inventory (heads)	0	56
Total beef inventory (ton)	5.2	55
Max. time beef held in inventory (month)	1	1

4.3.2. Trade-offs between multiple objectives

In our problem due to different features of trucks in terms of fixed renting cost and fuel consumption rates, trade-offs occur between logistics cost and amount of CO_2 emissions from transportation. This means that decreasing emissions from transportation comes at a cost. In addition to the aforementioned two base cases, we generated 10 additional instances by lowering ε value (limit on CO_2 emissions) 1% from the highest emission level at each instance. The derived Pareto frontier represents the trade-off relationships between cost and emissions (Fig. 5a).

As it is observed from Fig. 5a, the slope of the Pareto frontier is clearly decreasing after a point where emissions level is around 92%. The dotted horizontal line in Fig. 5a shows approximately the point where slope decrease starts. This slope change indicates that cost of achieving the same percentage of emissions reduction is increasing. This is because of the logistics network structure. In Nova-Andradina, the distances (average: 190 km) for the routes (3PL firms-production regions-slaughterhouses-3PL firms) in 1st stage are less than the distances (average: 1497 km) for the routes

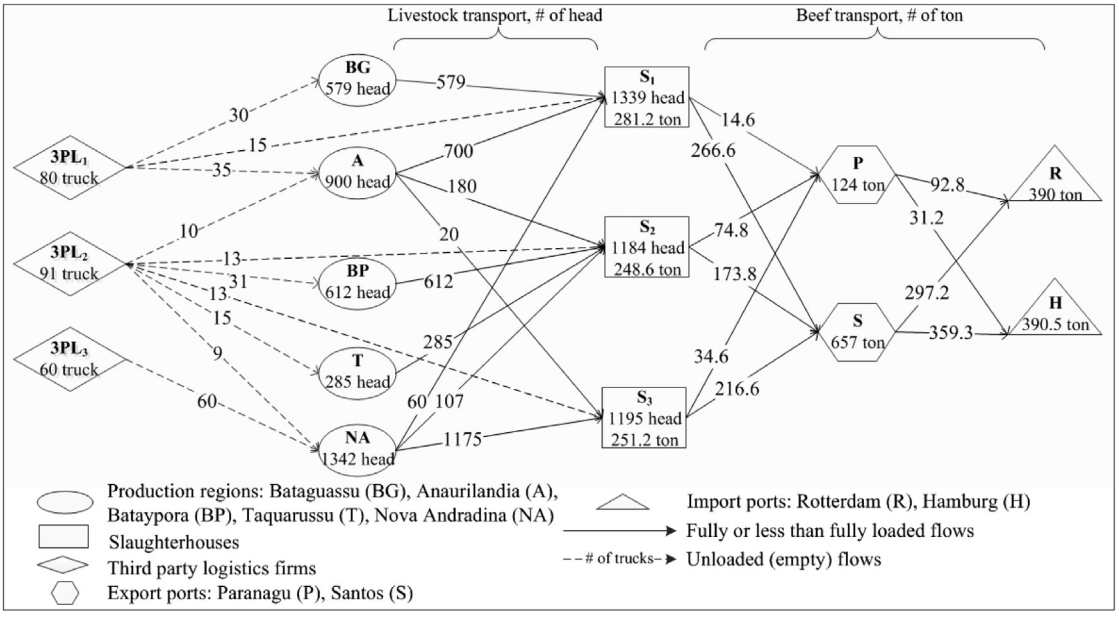


Fig. 4. Logistics network structure and related decision variables for the LC case.

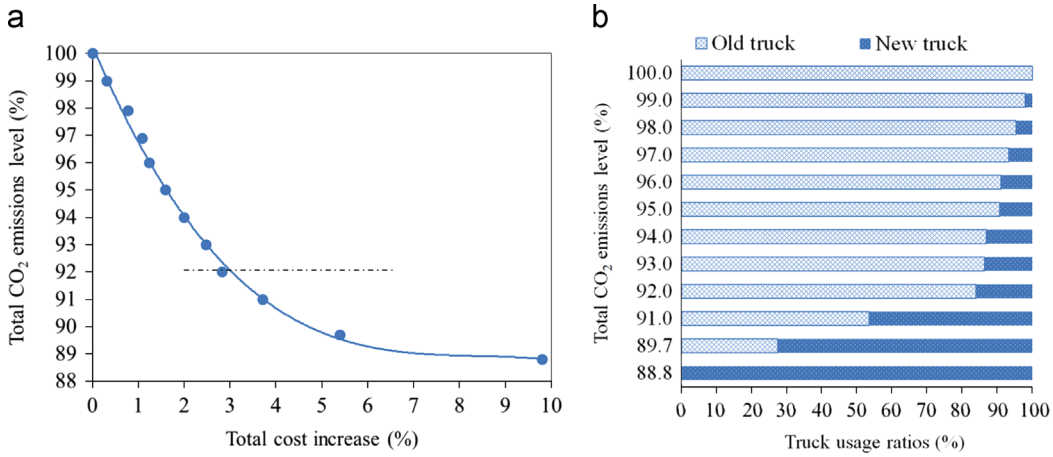


Fig. 5. Results obtained from the trade-offs analysis. (a) Trade-offs between total cost and CO_2 emissions. (b) Old and new truck usage ratios in all instances.

(3PL firms-slaughterhouses-export ports-3PL firms) in 2nd stage (see Fig. 3). Results show that new trucks are firstly rented for longer distances, in our case for the routes in 2nd stage, to reduce emissions. This is reasonable, since the total emissions effect of using a new truck will be higher for long distances compared to short distances. Model results suggest that emissions can be reduced by renting new trucks for the 1st stage after the trucks in the 2nd stage have been completely renewed. However, more new trucks need to be rented in the 1st stage to achieve the same emissions reduction, because of the distance differences between the 1st stage and the 2nd stage (see Fig. 3). This necessity results in higher cost increase for the same percentage of emissions decrease. Fig. 5b presents the truck usage rates in all instances. This figure also confirms that the ratio of new trucks is increasing more while achieving the same percentage of emissions decrease in further instances.

The trade-off analysis provides managerial insights on improving sustainability of the analysed logistics network. The cost of being sustainable from the point of reducing transportation emissions was determined by means of the Pareto frontier (see Fig. 5a) obtained through the trade-off analysis. This information is especially useful when setting sustainability targets that need an evaluation of economic and environmental factors. The Pareto frontier can be used for such an evaluation, since it ensures to find a compromise solution between costs and emissions. For instance, after evaluations of cost and emissions in line with economic and environmental objectives, one of the presented solutions in Fig. 5a can be selected for the analysed logistics network. One of the prominent sustainable suggestion would be selecting the breaking point located around the emissions level of 92% on the Pareto frontier (see Fig. 5a). This would ensure in approximate numbers an emission reduction of 8% in return of a cost increase of 3% that necessitates to use 16% new truck for the routes in 2nd stage (see Fig. 5a, b). As a result, these kinds of insights can support managerial decisions and improve sustainability performance of the selected chain.

4.3.3. *Road transport emissions ratios of different parts of the supply chain*

Considering all instances including the two base cases, average contributions of each supply chain part to total road transport emissions is presented in Fig. 6. More or less same amount of emissions are generated from ship transportation between export and import ports, which could raise the discussion of using local food. We know that trucks are either fully or less than fully loaded between production regions and slaughterhouses, and

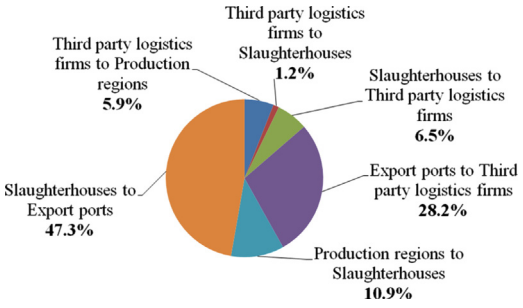


Fig. 6. Road transport emissions ratios in different parts of the supply chain.

slaughterhouses and export ports, and empty while coming from 3PL firms and turning back to them again in road transportation. Unsurprisingly, loaded trips (58.2%) contribute more than empty trips (41.8%) to total road transportation emissions and in loaded trips the share of slaughterhouses to export ports (47.3%) is higher than the share of production regions to slaughterhouses (10.9%) mainly due to the aforementioned distance differences. However, the remarkable thing is that empty trips constitute nearly two-fifths of emissions from road transportation. Among those empty trips especially the share of export ports to 3PL firms (28.2%) due to long distances is quite striking. Therefore, cost and emissions effect of decreasing empty legs (return hauls) between export ports and 3PL firms is significant to consider and will definitely improve the added value in logistics chain.

Addressing the more responsible parts for the road transport emissions in the whole chain helped us to point out some potential sustainable action plans. In our case, total emission contribution of the two parts, which are between slaughterhouses and export ports, and export ports and 3PL firms, is 75.5% due to the long distances (approximately 700 km) between these facilities. Managerial actions for dealing with such long distance operations in road transport might be as follows: (1) increasing the capacity of the vehicles, (2) using fuel efficient vehicles, (3) using alternative ports, (4) using backloading opportunities to reduce empty returns, and (5) shifting market from export to domestic market. Consequently, the managerial implication is that evaluating the road transport emission sources as a whole enables to present the importance of distances between actors in terms of environmental impact.

4.3.4. *Sensitivity analysis*

In order to get more insight, sensitivity analysis was conducted on five different parameters: export port capacities, fuel efficiency

levels, inventory holding costs, truck supply capacities and fixed renting cost of new trucks. Practical necessities and challenges learnt from the related literature and Brazilian partners in an EU funded project, SALSA, which has an objective of increasing social, environmental and economic sustainability of the Latin American and EU food chains (http://www.salsaproject.eu/, 2013), motivated us to focus on the aforementioned issues. Each of these analyses enables to explore the effects of possible changes in the current logistics system on cost and emissions. Therefore, the sensitivity analysis serves a supportive role in evaluating and better understanding the analysed beef logistics network in Brazil.

- *Effect of removing capacity restrictions on export ports:* The Porto de Paranagua is more closer (average: 660 km) to slaughterhouses and 3PL firms, than Santos Port (average: 780 km) in Nova Andradina region. However, capacity of the Porto de Paranagua allows for only 15.5% of export from Nova Andradina region to EU (Abiec, 2012c). We knew from the previous analysis that the total emissions share of slaughterhouses to export ports and export ports to 3PL firms constitutes 40.4% of emissions as well. These reasons led us to analyse the potential impact of removing capacity restrictions on the aforementioned export ports. As it is expected, removing capacity restrictions on export ports results in transportation of all European demand from the closer port, Porto de Paranagua. In this case, results show that a cost reduction of 2.0% in LC case, and 4.0% in LE case, and an emissions reduction of 4.2% in LC case, and 4.8% in LE case could be obtained. The satisfaction of all demand from Porto de Paranagua necessitates approximately a 6-fold capacity increase and because of financial issues it might not be possible within the short term. However, it is certain that improvements in port capacity of Porto de Paranagua would have major impacts on reducing cost and emissions in Brazil.
- *Effect of investing in better roads:* Transportation infrastructure has important implications on truck speeds and consequently on fuel efficiency (World-Bank, 2012). Brazil has the third largest road network in the world, at approximately 1.6 million km, but only 196,000 km (around 12%) of it is paved (World-Bank, 2010). This fact shows the inadequacy of road structure, or in general term transportation infrastructure, in Brazil. Correspondingly, actors in Brazil beef logistics chain pay taxes for the improvement of roads. However, we used the fuel consumption data for the trucks in Europe's conditions. This motivated us to analyse the impact of different fuel efficiency levels on both cost and emissions. We considered two fuel efficiency levels in which fuel consumptions were increased by 5% and 10% (Fig. 7). Results show that decreasing fuel efficiency due to bad infrastructure shifts the Pareto frontier to the right. This means that the trade-off relationship between the multiple objectives, costs and CO_2 emissions, is not affected, but both of them increase because of the increased fuel consumption. From the other perspective, results confirm that improvement by paying money for better roads has two potential awards: decreasing logistics cost and decreasing emissions. This sensitivity analysis also enables to evaluate the scenarios in which more energy is needed in Brazil for refrigeration, since the reason for the fuel increase might be not only bad infrastructure, but also excess energy usage of truck refrigerators in Brazil as well.

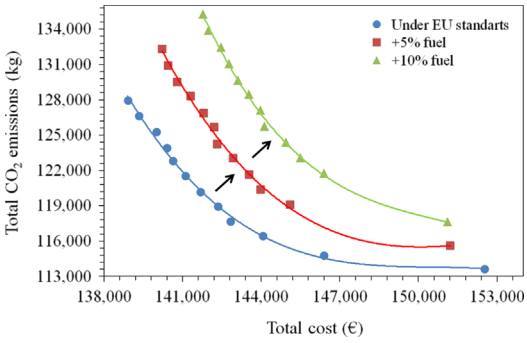


Fig. 7. Effect of increase in fuel consumption.

- *Effect of changing inventory holding cost:* Results on LC and LE cases (Table 3) show that beef is not held in inventory longer than 1 month, although maximum storage time is restricted to 2 months. However, handling of beef products that are 1 month old might be different for the wholesaler/retailer in terms of distribution and marketing processes required in comparison to that for fresh beef. This fact motivated us to analyse the effects of changes in inventory holding cost. In our previous analysis inventory holding cost for beef was expressed as a percentage, 3%, of the approximate market value of the product. Additionally, we considered 1% and 5% (Table 4). The lower inventory cost, 1%, ensures to analyse a scenario in which freshness of the product becomes less important and the higher inventory cost, 5%, ensures to analyse a scenario in which freshness of the product becomes more important. Results for the LC case show that increase in inventory holding cost leads to decrease in inventories and increase in road transportation cost. The reason for that is using less than fully loaded trucks and sending goods before aging become more advantageous rather than keeping items in inventory to increase load size in the next period. This can be observed from the percentages of less than fully loaded trucks for different inventory levels (Table 4). Moreover, due to more use of less than fully loaded trucks, total emissions increase as inventory holding cost increases. In summary, this sensitivity analysis enables to show the effect of changing inventory holding cost depending on the importance given on freshness of the beef.
- *Effect of restricting truck supply*: We assumed that whenever needed, trucks can be rented from the 3PL firms without any problem. However, the agricultural community in Brazil is concerned about truck shortages due to increased agricultural production and new regulations that restrict the number of hours a truck driver may work. Therefore, we analysed a scenario in which one of the three 3PL firms, $3PL_2$, is confronted with a truck shortage and cannot provide trucks during the whole planning horizon. Under this circumstance, facilities that rent trucks from $3PL_2$ necessarily go for the other two 3PL firms, $3PL_1$ and $3PL_3$, located comparably far. This supply change leads to increasing travelling distance that results in a cost increase of 1.1% in LC case, and 1.5% in LE case, and an emissions increase of 3.0% in LC case, and 2.7% in LE case. However, if the remaining two 3PL firms did not have enough truck capacity to compensate the supply decrease, trucks would be rented from 3PL firms located in more far away regions to satisfy the livestock and beef demand on time. The effect of that scenario on cost and emissions would be worse. Therefore, this sensitivity analysis on truck supply shows that capacity problems in 3PL firms have negative effects on cost and emissions.
- *Effect of changing fixed renting cost of new trucks*: We took fixed renting costs for the new truck higher than the old one considering related capital investment of 3PL firms depending on the age of the truck. One of the current discussions in logistics management is providing green tax incentives to encourage the purchase of cleaner vehicles consuming less fuel by means of new technologies (McKinnon et al., 2012). These kinds of legislations can reduce the related capital investment cost for the 3PL firms and the indirectly renting prices of new trucks. Therefore, we analysed the effects of reductions on fixed renting cost of new trucks (Table 5). According to the results for the LC case, 10% decrease does not affect on cost and emissions, since the model still suggests to use only old trucks in all routes as in the solution of LC

Table 4
Effect of changing inventory holding cost for the LC case.

Percentage of the market price	1%	3%	5%
Inventory cost (€)	789	546	37
Road transportation cost (€)	43,965	44,642	45,389
Total cost (€)	138,474	138,908	139,145
Total emissions (kg)	126,872	127,917	128,002
Less than fully loaded trucks (%)	8	12	13
Old truck usage (%)	100	100	100
Total livestock inventory (heads)	0	0	0
Total beef inventory (ton)	22.5	5.2	0.2
Maximum time beef held in inventory (month)	1	1	1

Table 5
Effect of changing fixed renting cost of new trucks for the LC case.

Percentage of decrease in rent (%)	Compared to LC case (%)		New truck usage (%)
	Cost reduction	Emission reduction	
10	0	0	0
20	0.1	3.5	9
30	1	7.5	18

base case (see Table 3). However, different amounts of reductions on cost and emissions are obtained for the instances where related costs are decreased by 20% and 30%. The reason for the emission decreases is starting to use new trucks. At first sight, cost increase would be expected for the last two instances, 20% and 30%, due to the usage of new trucks, which have still higher fixed renting costs compared to the costs for the old ones. However, less fuel consumption feature of new trucks makes use of them economically advantageous for the long routes even paying more for their rents. As a result, this sensitivity analysis on fixed renting cost of new trucks shows that green tax incentives can economically and environmentally improve the performance of the logistics systems.

5. Conclusions

In this study, we present a MOLP model for the generic beef logistics network problem. It has two competing goals: minimizing total logistics cost and minimizing total CO_2 emissions from transportation operations. Road structure, vehicle and fuel types, weight loads of vehicles, traveled distances, return hauls and product perishability are considered while integrating transportation emissions into the MOLP model. The model is thus important for decision makers who are concerned with logistical network problems of perishable products under emissions consideration. Implementation of the model on the beef export chain between a region of Brazil and the EU shows its applicability to real life logistics networks. The model can easily be further adapted to other emerging value-added food chains as well. The results presented in this study are obtained by means of a ε-constraint method used for solving MOLP models.

Different analyses are conducted to support decision making in the selected chain. The question of how much cost to bear to reduce emissions to different levels in the logistics system is answered by the presented Pareto frontier that shows the trade-off relationships between logistics costs and emissions. The next question is where to focus to (re)design a logistics network that is more environmentally friendly in terms of transportation emissions. Regarding that issue, road transport emission shares of the chain parts are observed by the pie chart. The pie chart results indicate the importance of distances between actors in terms of environmental impact. Moreover, sensitivity analysis on practically important parameters shows the potential impacts of several changes to logistics cost and emissions. Some of the most interesting results from the sensitivity analysis can be given as follows: (1) removing capacity constraints on export ports shows that capacities put pressure on the logistics system while selecting the related port for transportation, (2) decreasing fuel efficiency of trucks due to the inefficient infrastructure results in shifts of the Pareto frontier with both increase in logistics cost and emissions, and (3) decreasing fixed renting cost of new trucks due to the obtained advantage of 3PL firms from green tax incentives result in economic and environmental improvement. In summary, all the analyses show that the proposed model serves as a decision support tool while further improving the environmental position of the selected food logistics chain.

It is possible to extend the proposed model in several ways, which can be suggested as future research areas. First, reverse product flows from the destination nodes and indirect flows between facilities can also be considered. Second, other sources of emissions (e.g. emissions from livestock or refrigeration in more detail) and other sustainability key performance indicators (e.g. energy usage or water consumption) can also be evaluated. Third, quality or age of the products can be tracked through the supply chain in a more detailed way instead of just restricting the maximum number of periods that beef can be stored in facilities.

Acknowledgement

This research is part of the SALSA project that has received funding from the European Union Seventh Framework Programme (FP7/2007–2013) under Grant agreement number 265927.

Appendix A

In this section, we present the remaining data used for the MOLP model.

Table A1
Estimation of monthly European demand satisfied from Nova Andradina for fresh-chilled beef in the first 6 months of 2010, in tons.
Source: Abiec (2012a, 2012b) and Nzte (2010).

	Total beef export	EU share	Mato Grosso do Sul's participation	**Nova Andradina's participation (Demand)**
January	67,092	3355	394	**107**
February	74,311	3716	436	**119**
March	80,720	4036	474	**129**
April	79,010	3951	464	**126**
May	90,692	4535	532	**145**
June	96,856	4843	569	**155**
Total	953,869	47,693	5,599	**781**

Table A2
Distances between nodes in the beef logistics chain, in km.

	BG	A	BP	T	NA	S_1	S_2	S_3	P	S	$3PL_1$	$3PL_2$	$3PL_3$
Bataguassu (BG)	0												
Anaurilandia (A)	–[a]	0											
Bataypora (BP)	–	–	0										
Taquarussu (T)	–	–	–	0									
Nova Andradina (NA)	–	–	–	–	0								
S_1	42	39	66	99	45	0							
S_2	93	57	12	33	54	–	0						
S_3	93	66	39	54	30	–	–	0					
Porto de Paranagu (P)	–	–	–	–	–	639	640	702	0				
Santos Port (S)	–	–	–	–	–	740	772	825	–	0			
$3PL_1$	15	45	96	129	93	51	99	102	602	710	0		
$3PL_2$	66	48	15	45	45	57	15	24	644	774	–	0	
$3PL_3$	111	90	69	75	30	75	63	33	723	850	–	–	0

S_i: Slaughterhouse i, $3PL_i$: Third party logistics firm i.

[a] Distances which are not necessary for the model are not presented.

Table A3
Available livestock numbers in cities for each month, in heads.

	Jan.	Feb.	March	Apr.	May	June	Contribution[a] (%)
Bataguassu	88	97	107	105	122	130	16
Anaurilandia	132	145	161	158	183	196	24.1
Bataypora	85	94	104	102	118	127	15.6
Taquarussu	41	45	50	49	56	60	7.4
Nova Andradina	201	222	246	240	279	299	36.8
Total	547	603	668	654	758	812	100

[a] Contributions of cities in Nova Andradina to cattle production, in 2010 (Ibge, 2012).

Table A4
Summary for all the values and estimations of the problem's input parameters along with the relevant data sources.

Parameters	Values/estimations	Sources
liveinvcost	22.05€/head month	www.swineweb.com (2012), Cederberg et al. (2009)
beefinvcost	105)€/ton month	www.swineweb.com (2012), Cederberg et al. (2009)
rentcost	1st Stage: 40€/old truck, 70€/new truck 2nd Stage: 240€/old truck, 420€/new truck	Assumption
fuelcost	€1	http://data.worldbank.org (2012)
unitcost	0.12€/kg	www.globalshippingcosts.com (2012)
max	2 months	Assumption
weight	300 kg	Cederberg et al. (2009)
yield	0.7	Cederberg et al. (2009)
emptyfuel	Table 2	Hoen et al. (2010)
fullfuel	Table 2	Hoen et al. (2010)
conversion	2.63 kg/l	Defra (2007)
capacity	20 ton beef and 20 cattle	Cederberg et al. (2009)
distance	Table A2	www.maps.google.nl (2012); www.searates.com (2012)
demand	Table A1	Abiec (2012b), Nzte (2010),
livecap	Table A3	Ibge (2012)
liveinvcap	300 head	Assumption
slaughtercap	300 ton	Assumption
beefinvcap	Slaughterhouses: 200 ton, Export ports: 100 ton	Assumption
transcap	Porto de Paranagua: 124 tons, Santos Port: 676 tons	Abiec (2012c)
emissions	0.007 kg/ton km	Dekker et al. (2012)

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